

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

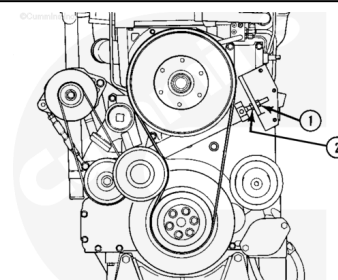
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the belt guards. Refer to Procedure 008-001 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-001-tr.html)
- Remove the fan and fan spacers. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-040.html)

Remove

⚠ WARNING ⚠

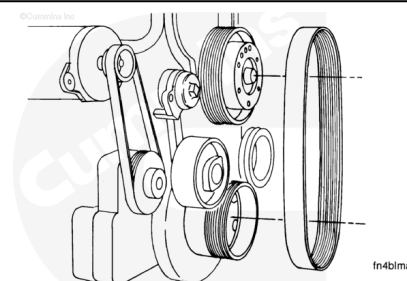
The fan belt idler is under tension. To reduce the possibility of personal injury, do not allow your hands to get between the idler and the belt, or the fan hub.



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Loosen the nuts (1 and 2). Turn the nut (1) **counterclockwise** to the end of the threaded rod to relieve the tension on the belt.

Remove the fan belt.



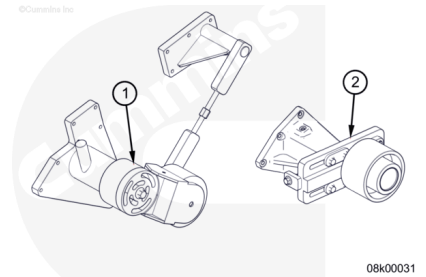
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Inspect for Reuse

Inspect the tensioner assembly (1) and idler pulley (2) for any cracks or excessive wear.

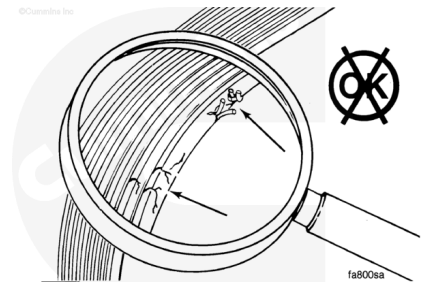
If the parts are cracked or worn, they **must** be replaced.

- Make sure the idler pulley moves freely
- Make sure bearing operation is smooth and does **not** have any free play
- Inspect the idler pulley for damage or wear.



Inspect the belt for:

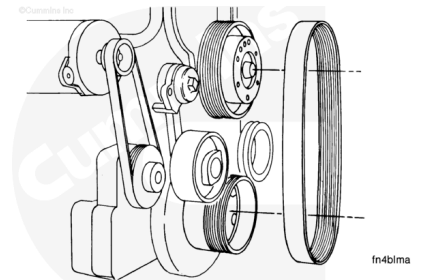
- Cracks
- Glazing
- Tears or cuts.



Install

Note : The fan belt has 21 grooves. The fan hub and crankshaft pulleys have 22 grooves. Install the belt in the grooves toward the front of the engine. The remaining groove that is **not** used is to be toward the radiator.

Install the belt.

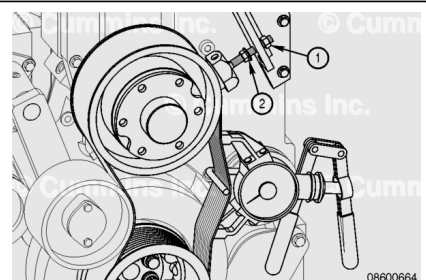


Adjust

Install the belt tension gauge, Part Number 2892563, or equivalent, on the right side of the belt, halfway between the two pulleys.

Loosen the inside nut (2) to the end of the threads.

Pull the belt tensioner until the idler pulley contacts the fan belt.



Tighten the nut (1) finger tight.

Tighten the outside nut (1) to a belt tension gauge reading of 1557 N [350 lbf].

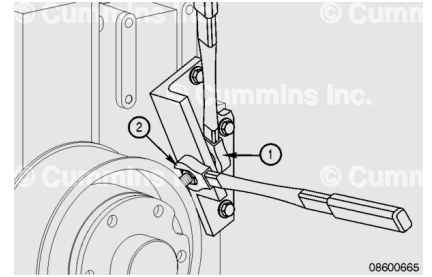
Finger tighten the inside nut (2) flush with the bracket before tightening.

Hold the outside nut (1) and torque the inside nut (2) to specification.

Torque Value: 81 n•m [60 ft-lb]

Bar the engine 360 degrees and recheck the belt tension.

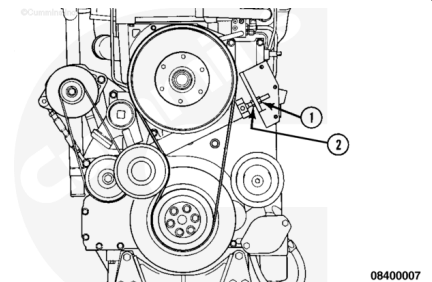
Make sure the tension is still within specification.



Note : If the belt was not replaced, it is not necessary to perform this step (belt removed and installed as part of another repair).

Operate the engine for 10 minutes at high idle. Reset the belt tension.

After this procedure, it is **not** necessary to tension the belt again until the belt is replaced.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the fan and fan spacers. Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-040.html)
- Install the belt guards. Refer to Procedure 008-001 in Section 8.
(/qs3/pubsys2/xml/en/procedures/20/20-008-001-tr.html)
- Operate the engine and check for noise.

